



IILM
UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA, UTTAR PRADESH

INTERNSHIP REPORT

ON

**DESIGN AND DEVELOPMENT OF A SAAS-BASED
SCHOOL MANAGEMENT SYSTEM USING MERN STACK**

Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Technology in Computer Science and Engineering

Submitted By

MD ASRAR

Roll Number: 2410031686

Section: 3CSE19

Semester: 5th

Batch: 2026–27

Internship Organization: Earthix Software Solutions Pvt. Ltd.

Internship Role: Full Stack Developer Intern

Internship Duration: 01 June 2026 – 31 July 2026

2026–27



INTERNSHIP OFFER LETTER

Candidate Name:	Md Asrar	Employee ID:	EX321151
Internship Role:	Full Stack Developer Intern	Joining Date:	01/06/2026
Duration:	2 Months	Work Mode:	Work From Home
Stipend:	Unpaid	Contact Number:	+91 8340452870

Dear Md Asrar,

We are pleased to offer you the position of **Full Stack Developer Intern** at **Earthix Software Solutions PVT. LTD.**

This internship is designed to provide practical exposure, structured learning, mentorship, and hands-on project experience in Full Stack Web Development within a professional work environment.

During your internship, you will collaborate with the Development, Design, and QA teams, work on assigned full stack deliverables, and gain valuable industry experience in both frontend and backend technologies.

Internship Benefits

- Hands-on practical learning in full stack development (MERN / MEAN Stack)
- Real-world project experience on live website and web applications
- Internship completion certificate
- Letter of recommendation based on performance
- Mentorship from senior full stack developers
- Future employment consideration based on performance and openings

Roles & Responsibilities

- **Frontend Development:** Develop and maintain responsive, user-friendly web interfaces using HTML, CSS, JavaScript, and modern frameworks like React.js / Next.js.
- **Backend Development:** Build and manage server-side logic, APIs, and database operations using Node.js, Express.js, and related backend technologies.
- **Database Management:** Design, implement, and maintain databases (MongoDB / MySQL / PostgreSQL), ensuring data integrity, security, and optimized performance.
- **API Development & Integration:** Create and integrate RESTful APIs, handle third-party API integrations, and ensure seamless communication between frontend and backend.
- **Full Stack Project Execution:** Work on end-to-end development of live projects, from requirement analysis and UI implementation to deployment and maintenance.
- **Version Control & Collaboration:** Use Git/GitHub for version control, collaborate with cross-functional teams, participate in code reviews, and follow agile development practices.
- **Bug Fixing & Optimization:** Debug, test, and optimize applications for maximum speed, scalability, and security across different browsers and devices.
- **Professional Conduct:** Adhere to company policies, maintain confidentiality of project data and source code, meet deadlines, and participate actively in team meetings and reviews.

Terms & Conditions

1. Internship duration is from **1st June 2026 to 31st July 2026** and will continue for 2 Months.
2. This is a Work From Home internship. Intern must have a stable internet connection and laptop/desktop for development activities.
3. All work products, source code, designs, and documentation created during the internship are the intellectual property of Earthix Software Solutions Pvt. Ltd.
4. Confidentiality of all company data, client information, systems, and intellectual property must be strictly maintained even after internship completion.
5. Intern must maintain minimum 90% attendance and submit weekly progress reports to the reporting manager.
6. Unsatisfactory performance, breach of confidentiality, or misconduct may result in immediate termination of internship without certificate.
7. Completion certificate and letter of recommendation are subject to satisfactory performance and completion of assigned tasks.
8. This internship does not constitute an offer of permanent employment, though exceptional performers may be considered for full-time roles.

Acceptance & Confirmation

Candidate Signature: _____

Candidate Name: Md Asrar

Acceptance Date: _____

Earthix Software Solutions Pvt. Ltd.

Authorized Signatory: Earthix Software Solutions Pvt. Ltd.

Ph : 9217232660
www.earthix.in

For more queries, visit our website: www.earthix.in

✉ support@earthix.in

Full Stack Development Internship

Earthix Software Solutions Pvt. Ltd.

THIS CERTIFICATE IS PROUDLY PRESENTED TO

MD ASRAR

For Successfully Completing the

Full Stack Developer Internship

This certificate is awarded in recognition of the successful completion of the **Full Stack Developer Internship** at **Earthix Software Solutions Pvt. Ltd.**

The internship was successfully completed from **01 June 2026** to **31 July 2026** for a duration of **Two Months (8 Weeks)**.

During the internship period, the intern demonstrated excellent technical skills, dedication, problem-solving ability, and a strong willingness to learn while making valuable contributions to the full stack development projects including frontend, backend, API development and database management.

Employee ID : EX321151

Issued Date : 01/08/2026

Certificate ID : ESSPT321151

Earthix Software Solutions Pvt. Ltd.
Md Sanaullah
Director

Earthix Software Solutions Pvt. Ltd.

Earthix Software Solutions Pvt. Ltd.
Ph : 9217232660

www.earthix.in
Earthix Software Solutions Pvt. Ltd.

1. CANDIDATE’S DECLARATION

I, MD ASRAR, Roll Number 2410031686, student of B.Tech Computer Science and Engineering, Section 3CSE19, Semester 5th, hereby declare that this internship report titled “Design and Development of a SaaS-Based School Management System using MERN Stack” has been prepared for academic submission to IILM University in connection with my internship at Earthix Software Solutions Pvt. Ltd.

I declare that the report has been prepared for academic purposes and that the supporting internship documents included at the end of this report are the documents issued/provided for my internship. I understand that I am responsible for maintaining academic integrity and for appropriately acknowledging material used in this report.

Signature: _____

Student Name: MD ASRAR

Roll No.: 2410031686

2. ACKNOWLEDGEMENT

I sincerely express my gratitude to Earthix Software Solutions Pvt. Ltd. for providing me the opportunity to undertake a Full Stack Developer Internship and gain practical exposure to web application development. The internship provided an opportunity to understand professional development practices and the workflow involved in building and maintaining full stack applications.

I am thankful to the mentors and team members who supported me with technical guidance, feedback and practical learning during the internship. The experience helped me strengthen my understanding of frontend development, backend development, RESTful APIs, database operations, version control, debugging and application optimization.

I also express my sincere thanks to the faculty members and internship coordinators of the School of Computer Science and Engineering, IILM University, for their academic support and guidance. Finally, I am grateful to my family and everyone who encouraged me throughout the internship and helped me complete this report.

3. INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by Earthix Software Solutions Pvt. Ltd. is reproduced in its original visual form in the supporting-documents section of this report. The certificate states that MD ASRAR successfully completed the Full Stack Developer Internship from 01 June 2025 to 31 July 2025 for a duration of two months (8 weeks).

4. PROJECT DESCRIPTION

4.1 Introduction

The internship project is presented as the design and development of a SaaS-based School Management System using the MERN stack. The proposed platform is intended to centralize common school administration and academic activities in a web-based environment.

Software as a Service (SaaS) enables users to access software through the internet rather than maintaining separate installations. In a school-management context, a SaaS platform can provide centralized data management, role-based access and a scalable foundation for supporting multiple schools.

The system is designed around multiple user roles such as administrators, teachers, students and parents. It brings functions such as student records, attendance, fees, examinations, timetable management and communication into a unified application.

4.2 Organization Profile

Earthix Software Solutions Pvt. Ltd. was the organization where the internship was undertaken. The internship offer letter identifies the role as Full Stack Developer Intern and states that the internship was intended to provide practical exposure, structured learning, mentorship and hands-on project experience in Full Stack Web Development.

The offer letter lists responsibilities including responsive frontend development, backend logic and APIs, database operations, RESTful API development and integration, end-to-end project execution, Git/GitHub collaboration, debugging, testing and optimization. The internship was conducted in Work From Home mode for two months from 01 June 2025 to 31 July 2025.

4.3 Problem Statement

Schools manage substantial information related to students, teachers, attendance, fees, examinations, timetables and communication. When these activities are maintained through disconnected records or separate tools, information can become difficult to access, update and manage consistently.

The project therefore addresses the need for a centralized web-based management platform that organizes important school processes and provides appropriate role-based access to different users.

4.4 Project Objectives

- Develop a centralized web-based platform for school management.
- Provide role-based access for administrators, teachers, students and parents.
- Organize student and teacher information in a structured manner.
- Support attendance, fee, examination and timetable management.
- Provide RESTful APIs for communication between frontend and backend.
- Use a document-oriented database for structured application data.
- Apply Git/GitHub-based version control and collaborative development practices.
- Create a scalable foundation that can be extended with additional SaaS modules.

4.5 Scope of the Project

The proposed system covers major academic and administrative functions of a school. An administrator can manage users and institutional information; teachers can work with attendance and academic activities; students can view relevant academic information; and parents can monitor student-related information.

The platform can be extended with online fee payment, notifications, report generation, analytics, mobile applications and integrations with external educational services. A SaaS-oriented architecture can also support multiple schools with appropriate separation of each institution's data.

4.6 Technologies and Tools Used

Category	Technology / Tool	Purpose
Frontend	HTML, CSS, JavaScript, React.js	User interface and responsive web pages
Backend	Node.js, Express.js	Server-side logic and REST APIs
Database	MongoDB	Application data storage
API	RESTful APIs	Frontend-backend communication
Version Control	Git, GitHub	Source control and collaboration
Development	Visual Studio Code	Development environment

Testing	Postman	API testing and verification
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4.7 System Architecture

The proposed application follows a three-layer web architecture. The React.js frontend provides the user interface and communicates with the backend through RESTful APIs. The Node.js and Express.js backend handles request processing, business logic, validation and API services. MongoDB acts as the database layer for storing application information.

A typical request begins at the user interface, moves to an API endpoint on the backend, is validated and processed, and then interacts with the database when data needs to be created, retrieved or updated. The processed response is returned to the frontend for presentation to the user.

4.8 Methodology

- Requirement Analysis – identify users, modules and functional requirements.
- System Design – plan application structure, roles, data entities, API flow and interface.
- Frontend Development – build reusable React components and responsive screens.
- Backend Development – implement Node.js/Express.js services and REST API endpoints.
- Database Development – design MongoDB collections and required data structures.
- Integration – connect frontend screens with backend APIs and verify data flow.
- Testing and Debugging – test workflows and APIs, identify defects and improve reliability.
- Version Control – manage source code using Git/GitHub and organized development practices.

4.9 Expected Outcomes

The expected outcome is a structured web-based school management platform that centralizes important information and workflows. The system is intended to improve

organization, accessibility and efficiency while providing a scalable base for future SaaS expansion.

The internship also provides practical learning in full stack development, component-based frontend design, backend API development, database management, version control, testing and debugging.

4.10 Certificates of Completion and Communication Proof

The supporting-document section of this report contains the original visual pages of the Earthix Internship Completion Certificate and Internship Offer Letter. They are inserted as page images so that their wording, layout, signatures, logos and other visual elements remain unchanged from the supplied documents.

The offer letter identifies the internship as a Full Stack Developer Intern position, with a two-month duration and Work From Home mode. The completion certificate records successful completion of the Full Stack Developer Internship from 01 June 2025 to 31 July 2025.

5. BIBLIOGRAPHY / REFERENCES

1. Earthix Software Solutions Pvt. Ltd., Internship Offer Letter – Full Stack Developer Intern.
2. Earthix Software Solutions Pvt. Ltd., Certificate of Completion – Full Stack Development Internship.
3. IILM University, Internship Report Format updated 26-27.
4. Documentation and learning resources used during the internship, where applicable.